An Intersectional Framework for Explaining and Mitigating Bias in Automated Hiring

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1 Introduction

Recently, artificial intelligence (AI) has become an integral component of decision-making systems across different domains. They are used in credit scoring, predictive policing, healthcare, and employment. Among all of these, automated hiring systems represent one of the most socially critical areas of AI use. By evaluating candidates, these systems promise efficiency, cost reduction, and fairness. However, growing evidence shows that these tools may accidentally reproduce and even strengthen historical patterns of discrimination, especially in terms of gender, ethnicity, and age. Crucially, these biases often do not act in isolation but rather interact to form barriers for intersectional groups such as older women. This is known as intersectional discrimination [Crenshaw, 1989]

This tension between technological efficiency and social justice creates serious ethical and legal questions. If a hiring system denies a job opportunity, who is responsible for that? The developer, the employer, or the dataset? Furthermore, how can we, the humans, challenge or understand decisions that have been made by machine learning systems?

At the core of this issue lies the phenomenon of algorithmic bias, meaning the systematic and unfair disadvantage toward specific social groups encoded in AI systems. Bias can occur during data collection, feature selection, model training, or inference and may continue to show even when sensitive attributes such as gender or race are explicitly excluded [Barocas and Selbst, 2016]. This discrimination not only harms people's dignity, but it also violates fundamental human rights principles, including the right to equality and fair treatment in employment [Assembly, 1949].

In response to these challenges, Explainable AI (XAI) has emerged as a crucial approach, trying to make complex models and their decisions more transparent and interpretable. Methods such as Shapley Additive Explanations (SHAP) enable to observe which input features most strongly influence a models output. However, diagnosis alone is insufficient. To move from observing bias to correcting it, XAI must be integrated with active fairness frameworks. Combined with tools such as Fairlearn, these methods provide an empirical foundation for assessing, visualizing, and actively mitigating bias in automated decision making.

This paper should explore the relationship between technical transparency and human rights protection. It moves beyond simple diagnosis to propose an effective auditing pipeline. It examines how bias manifests in automated hiring algorithms, putting special focus on intersectionality, and investigates how XAI and fairness tools can be combined to identify, explain, and effectively remove discriminatory outcomes.

2 Theoretical Frameworks

2.1 The Illusion of Neutrality in Automated Decision-Making

Automated decision-making systems are increasingly used in recruitment processes to screen applicants or predict job performance. These tools are often perceived as more objective and efficient than traditional human decision-making. However, recent studies show that such systems frequently reproduce or even amplify existing social inequalities [Barocas and Selbst, 2016]. With the knowledge that data reflects historical power structures, we realize that algorithms are not neutral. Rather, they often function as black-boxes that hide systematic discrimination behind seemingly mathematical objectivity. When models are trained on historical hiring data, they do not just learn who is best suited for a specific role. They rather learn who has historically been selected, therefore encoding societal biases regarding gender, race, or age directly into their decision logic.

2.2 What is Fairness?

In this context, algorithmic fairness has emerged as an important concept linking technical design with ethical and legal principles. In computer science, fairness usually refers to the equal treatment or outcomes across different demographic groups, measured by metrics such as demographic parity or equal opportunity [Dwork et al., 2012]. In human rights, fairness corresponds to the principles of equality, justice, and lack of discrimination. These values are deeply rooted in international law, most prominently in the Universal Declaration of Human Rights (UDHR) [Assembly, 1949] and the European Convention on Human Rights [of Human Rights, 2010], which states that all individuals have the right to equal treatment and participation in employment. Translating such fundamental principles into mathematical constraints, however, remains one of the most difficult challenges in AI [Binns, 2018].

2.3 The Problem with Single-Dimensional Discrimination

A major limitation of current theoretical frameworks is the focus on single dimensional discrimination. Traditional audits often evaluate fairness regarding gender or age in isolation. This approach overlooks the theory of intersectionality [Crenshaw, 1989] Intersectionality is the idea that people often experience multiple forms of discrimination, such as the overlap of sexism and ageism.

In algorithmic auditing, ignoring intersectionality leads to a phenomenon known as fairness gerrymandering [Kearns et al., 2018]. A hiring model might appear statistically fair to women as a group by hiring a disproportionate number of young women, while systematically excluding older women. If an audit only checks for gender or age bias separately, this specific subgroup discrimination remains invisible. Therefore, novel frameworks for automated decision systems must go beyond isolated bias analysis to detect intersectional discrimination.

2.4 The Role of Explainability and the Right to Explanation

The field of Explainable Artificial Intelligence (XAI) attempts to address part of this challenge by making the outcome of machine learning models transparent. Methods such as SHAP [Lundberg and Lee, 2017] quantifies the contribution of each input variable to the models final predictions. Explainability helps developers and users understand why a model made a certain

decision and where bias might occur. It connects technical transparency with the legal right to a fair process, as described in Article 10 of the Universal Declaration of Human Rights and the General Data Protection Regulations “right to explanation” [Goodman and Flaxman, 2017]. Without explainability, an applicant can not challenge a discriminatory automated hiring decision. Consequently, XAI serves not just a technical function, but also acts as a prerequisite for legal accountability.

2.5 From Ethics to Regulation

New regulations are beginning to address the risks of opaque AI systems, shifting from voluntary ethical principles to binding legal rules that focus on fairness and accountability [Floridi et al., 2018]. Most notably, the European Union AI Act [Commission, 2024] classifies AI systems used in employment and recruitment as "high risk". This classification raises the need for high standards of data quality to prevent discriminatory outcomes.

Ethical and legal principles must now be supported by real technical methods. Algorithmic fairness and explainability provide a foundation for creating AI systems that protect human rights by checking equality and making bias visible. However, diagnosis alone is not enough. Compliance with the new legal standards requires active mitigation strategies that can address the intersectional nature of real-world discrimination.

2.6 Alignment with Horizon Europe Principles

This research proposal is aligned with the objectives of Horizon Europe, specifically under Cluster4: Digital, Industry & Space [European Parliament and Council, 2021]. The European Commission aims to assist more compact, more efficient and less energy intensive, as well as human-centric and trustworthy AI.

Horizon Europe prioritizes AI that serves humanity. Our focus on intersectional fairness directly addresses the priority of gender equality and inclusiveness. By developing tools that protect vulnerable subgroups from algorithmic exclusion, this project contributes to the human-centric ai, ensuring that the digital transformation remains accessible to all demographics.

Furthermore, a key objective of Horizon Europe is ensuring that European AI is robust and accountable. Currently, European companies frequently rely on black-box solutions, often imported from non-European organizations. By proposing an open-source auditing framework, this proposal strengthens digital sovereignty. It provides European regulators and civil society with independent tools to verify compliance with the EU AI Act, thereby reducing dependency on the internal reporting mechanism of large technology providers. Consistent with the Horizon Europe open science policy, this proposal requests that all developed auditing pipelines be released as open-source software, bridging the gap between scientific innovation and societal application.

3 Mechanisms

3.1 Historical Data and Intersectional Patterns

To understand how bias appears in automated hiring, it is important to look at how human decisions, data collection, and algorithms interact. Training data often reflect existing

inequalities in the job market, such as gender roles, racial discrimination, or differences in income and education. If women or minority candidates have historically been hired less often, an algorithm trained on this data will learn those patterns and treat them as normal. As a result, the system can continue to make unfair hiring decisions, even if no one involved intends to discriminate. However, a deeper analysis reveals that these biases are rarely dependent on isolated groups. Intersectional theory suggests that data often encodes combinations of traits such as gender and age where the historical hiring rate for specific subgroups is significantly lower than for the individual groups alone.

3.2 Automation Bias and Human Oversight

Another factor is automation bias, which means that people tend to trust the output of algorithms too much [Goddard et al., 2012]. Recruiters may rely on automated recommendations simply because they appear objective or data-driven. An overreliance can make existing bias even stronger, as the algorithms results are rarely questioned. This is particularly dangerous in the context of intersectional discrimination. A recruiter might manually verify that the candidate list contains enough women and enough older people to appear fair, failing to notice the systematic exclusion of older women due to trusting the aggregated system output.

3.3 Proxies and Cumulative Disadvantage

From a technical point of view, machine learning models find patterns and correlations in data. Variables such as education level or working hours can act as proxies for sensitive information, such as gender, ethnicity or age. Even if these sensitive features are removed, the model can still infer them through related data. Beyond proxies, an important mechanism is the concept of cumulative disadvantage [DiPrete and Eirich, 2006]. A machine learning model may learn to assign a small negative weight to being female and a small negative weight to being older. Individually, these penalties might not be enough to lead to a rejection, however, for a candidate who belongs to both groups, these penalties accumulate, pushing the candidate below the hiring threshold. XAI methods help uncover these hidden relationships by showing which features influence the models decisions most.

3.4 The Black-Box Problem

Finally, from a legal and ethical perspective, there are often no clear rules for auditing these systems. Many commercial hiring tools are proprietary and not open to the public, making it difficult for rejected applicants to understand why a decision was made. XAI can address the issue of lack of transparency by demonstrating how algorithmic decisions relate to legally protected characteristics.

4 Consequences

The consequences of bias in automated hiring systems are significant, ranging from individual discrimination to structural inequality [Zajko, 2022]. As these systems become gatekeepers to economic opportunity, their errors do not merely result in technical inaccuracies but in the systematic exclusion of entire demographic groups.

4.1 Structural Inequality

For women, machine learning models that rely on biased historical data can maintain gender gaps in leadership and technical positions. Historic hiring data often reflects a reality where men dominated specific sectors [Barocas and Selbst, 2016]. Algorithms trained on this biased data, learn to penalize women. Amazon's automated CV screening tool, which systematically downgraded female candidates for engineering roles, is a good example of how bias can manifest without explicit sexist intent [Dastin, 2022]. The system penalized resumes containing the word "women's" simply because such terms were statistically less frequent among successful past hires. Such systems undermine the principle of equal access to employment opportunities, directly contravening Article 23 of the UDHR.

4.2 Intersectionality

While gender and racial biases are well documented, a more harmful consequence emerges at the intersections of these features. Ethnic and racial minorities are often affected by proxy variables such as zip code or educational background which encode socioeconomic disparities. However, the most critical consequences occur when multiple disadvantages add up.

This proposal specifically highlights the challenges of the intersectional subgroup of older women. Age bias is a problem where algorithms implicitly penalize older or very young candidates. However, when age bias interacts with gender bias, it creates a cumulative disadvantage. An older male might be hired for his experience and a young female might be hired for diversity targets. But older female candidates face the negative weights of both ageism and sexism. The consequence is not just a lower probability of hiring, but often total exclusion [Kearns et al., 2018]. As we will show in our subsequent experiment, standard auditing tools that look at women and older people individually, may report acceptable fairness metrics, while the intersectional subgroup faces a hiring rate of near zero. This invisible discrimination is the most dangerous consequence of automated hiring as it leaves rejected candidates from these subgroups no opportunity to question the hiring decision because the statistic appears fair.

4.3 Loss of Trust

In addition to these measurable outcomes, algorithmic discrimination leads to broader social risks. It destroys public trust in AI, discourages individuals from applying for positions mediated by algorithms, and strengthens existing social hierarchies in a seemingly neutral way. Furthermore, the opacity of these systems contributes to automation bias among recruiters. Because the output is a mathematical score, human decision-makers are less likely to question it than they would a human recommendation. This creates a feedback loop where the algorithms intersectional biases are accepted as objective assessments of candidate skill, permanently locking vulnerable minorities out of the job market.

4.4 Algorithmic Anxiety and Learned Helplessness

Beyond the structural exclusion form the job market, opaque hiring algorithms create distinct psychological stress for applicants [Hilliard et al., 2022]. This phenomenon is known as algorithmic anxiety [Elliott, 2024]. Candidates often modify their behavior not to demonstrate

competence to a human reader, but to satisfy the perceived parameters of an automated filtering system. This even leads to adaptive behaviors such as keyword stuffing, where applicants hide white text in margins or repeat certain keywords, degrading the quality of communication in the hiring process.

Furthermore, rejection by a black-box system without feedback can create a sense of helplessness. Unlike human rejection, which can sometimes be contextualized or reasoned with, algorithmic rejection appears objective and unchangeable. For marginalized groups who already experience imposter syndrome [Breeze, 2018], this neutral rejection can lead to withdrawal from the job market entirely, as the barrier to entry is perceived as too high.

4.5 Economic and Organizational Consequences

Finally, bias must be understood as an economic responsibility for the organizations deploying automated tools. Fairness gerrymandering [Kearns et al., 2018] results in false negatives, systematic rejection of highly qualified candidates because they belong to an intersectional subgroup, such as experienced older women. In a competitive job market, companies relying on biased algorithms act against their own financial interests by artificially reducing their available talent opportunities.

Furthermore, under the requirements introduced by the EU AI Act, the operational risk of deploying discriminatory systems has increased. High risk systems that are found to not follow the requirements face significant financial penalties [Commission, 2024]. Consequently, the persistence of intersectional bias is no longer just an ethical concern but a real compliance risk that threatens the long term sustainability of automated recruitment strategies.

5 Models and Experiment

To answer the critical question of how algorithmic bias can be not only detected but effectively removed, we conducted an experiment. We implemented a transparent Logistic Regression framework. This choice aligns with the "right to explanation" and focuses on interpretability over a black-box approach in important hiring scenarios. This chapter describes our novel auditing pipeline, which integrates XAI using SHAP for diagnosis and Intersectional Fairness Optimization with Fairlearn for mitigation.

5.1 Methodology

To test the efficacy of bias mitigation tools, it is insufficient to rely on datasets where the source of bias is unknown. Instead, we use a simulated approach. We employed a recruitment dataset [El Kharoua, 2024] consisting of standardized applicant profiles with features like *Education Level*, *Age*, *Gender*, and *SkillScore*.

5.1.1 Simulation of Intersectional Bias

We introduced a synthetic bias into the training data to especially highlight the group of older women. This group is an intersection, effected by both ageism and sexism. Unlike simple bias (only one group is rejected), we implemented a "Double Penalty" logic:

- **Baseline:** The hiring decision was originally mostly based on an applicants *SkillScore*.

- **Bias injection:** We applied a penalty to the skillscores of specific demographic groups using the following function:

$$SkillScore_{biased} = SkillScore_{original} - (\alpha \times I_{Gender=Female}) - (\beta \times I_{Age>45}),$$

where $\alpha, \beta = 10$.

- **Outcome:** Candidates were marked as hired only if their penalized score was over a threshold of 70.

This approach created a scenario of disadvantage. While a penalty of 10 points might be negligible for a young woman or an older man, an older woman faces a sum of 20 penalty points. This makes it nearly impossible for this subgroup to be hired, resulting in candidates from this group being systematically excluded.

5.1.2 Model Architecture

We trained a Logistic Regression classifier on this biased dataset. The model achieved a total accuracy of 0.95. This score deceptively suggests high performance while hiding the critical discrimination against the intersectional subgroup.

5.2 Model Diagnosis

The first step in our pipeline addresses the explainability of model decisions. Standard fairness metrics can indicate that some sort of bias exists, but they do not explain the models reasoning. Therefore, we used SHAP. As shown in Figure 1, SHAP successfully "audited" the models

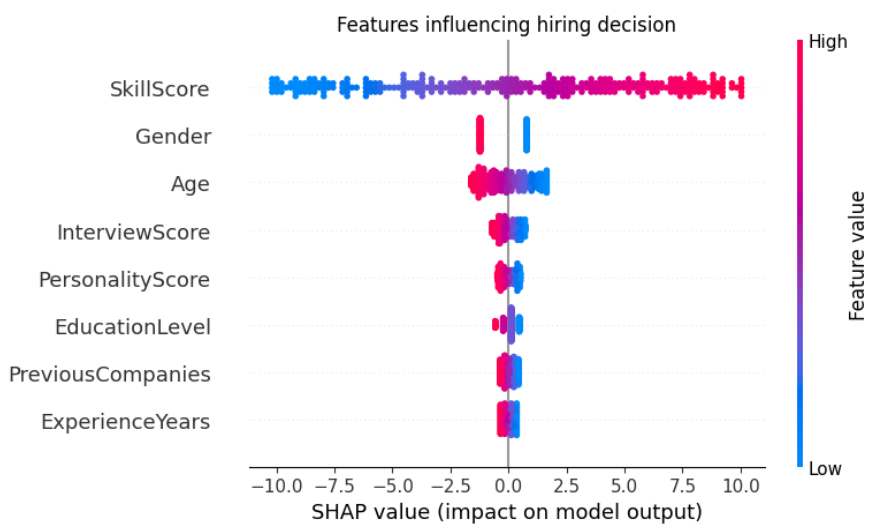

Figure 1: Visualised SHAP values of the features influencing the models' hiring decision. The sensitive variables gender and age are very important for the decision.

internal logic. The summary plot reveals that the *SkillScore* is still the strongest positive predictor for hiring. However, the plot visualizes that *Gender* and *Age* possess significant negative Shapley values. This visual evidence provides the necessary proof for auditors that the model is not acting neutrally but is explicitly utilizing sensitive attributes to penalize

candidates. Figure 2 gives an even more detailed look at some of the bias in our model. Unfortunately, diagnosis is not a cure. While SHAP confirms the mechanism of the bias, it does not remove it. This limits the utility of XAI as a standalone solution for human rights protection.

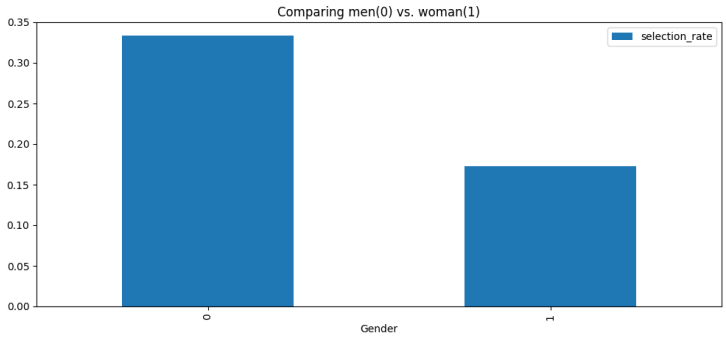

Figure 2: Hiring rate between male (left) and female (right) candidates. A clear gender bias is visible.

5.3 The Problem with Standard Auditing Practices

A key finding in our experiment is the insufficiency of standard auditing practices. When we calculated the hiring rate for single demographic groups, the results show a problem:

- Hiring Rate (all woman): $\approx 18\%$
- Hiring Rate (all old people): $\approx 15\%$

However, when we analyzed the intersectional subgroup (woman aged < 45), the hiring rate dropped to 0. This allows a model to appear statistically acceptable by hiring enough young woman to satisfy gender quotas, while completely excluding the group of older women. This confirms that standard Fairlearn metrics, due to the lack of intersectionality, fail to protect human rights.

5.4 Intersectional Mitigation

To address this problem, we implemented a Post-Processing mitigation strategy using the *ThresholdOptimizer* from the Fairlearn library.

5.4.1 Combined Sensitive Features

Standard mitigation often constraints models to achieve demographic parity for gender or age. Our approach creates a composite sensitive feature:

$$Feature_{sens} = Gender \bigoplus AgeGroup$$

This forces the optimizer to treat "young male, "old male", "young female", and "old female" as distinct groups that must each satisfy the fairness constraints.

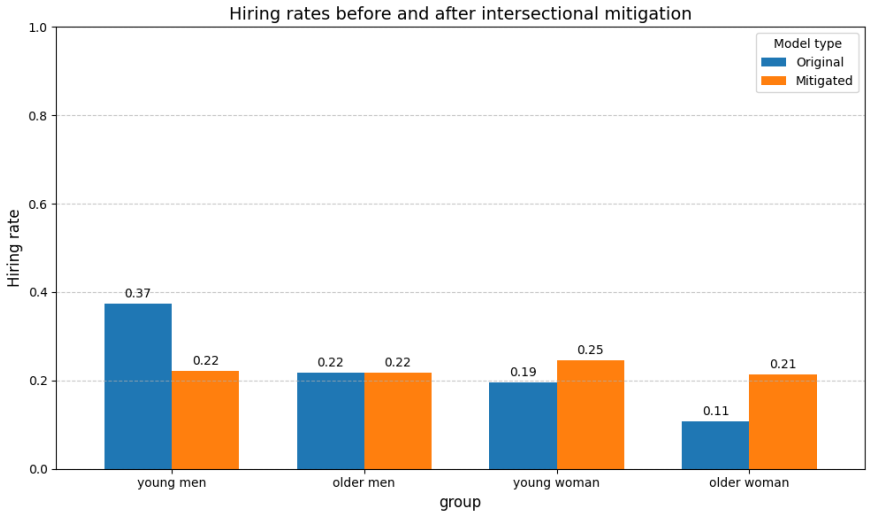

Figure 3: Hiring rates before (blue) and after (orange) mitigation for the different intersectional subgroups. The *ThresholdOptimizer* was able to mitigate the bias.

5.4.2 Results

The *ThresholdOptimizer* recalculates the classification decision boundary for each subgroup independently. It does not change the models weights but rather adjusts the acceptance standard to compensate for the introduced bias using the "Double Penalty" approach explained before. We can see in Figure 3 that the mitigation attempt was successful.

5.4.3 The Cost of Fairness

This restoration of demographic parity came with a small trade-off in model performance. The models accuracy dropped from 92 to 90 which demonstrates that even after our pipeline, the model is still usable.

5.5 Experiment Conclusion

This experiment provides a good proposed addition to modern auditing. By combining SHAP with intersectional Fairlearn, we developed a new auditing pipeline. This approach goes beyond mere observation, but actively restores the human right to equal treatment for marginalized subgroups. The code used for this experiment can be found in https://github.com/Tjaardo/Intersectional-Bias-in-Automated-Hiring.git

6 Proposed Research Agenda

While the experiment in the previous chapter 5 demonstrated the technical feasibility of intersectional bias mitigation on structured data, a full-scale implementation requires a significantly broader scope. Our model relied on simplified tabular data, however, modern recruitment processes are far more complex and unstructured. To translate our initial findings into a deployable auditing framework, this research proposal outlines a project structured over the course of 24 months. We structured it into three distinct work Packages.

6.1 Work Package 1: Analysis of Unstructured Data

The primary limitation of our study was its reliance on pre-structured features such as *SkillScore* or *ExperienceYears*. In real recruitment, however, most of the applicant data exists in unstructured formats, primarily CVs and cover letters. Logically, the first phase of this research will extend the auditing pipeline to Natural Language Processing (NLP).

We propose to investigate how Large Language Models (LLMs) used in automated hiring systems encode intersectional bias within their text embeddings. A central research question is whether specific semantic patterns correlate with intersectional identities. As an example, we could analyze whether language and their respective embeddings often associated with older women yields lower similarity scores to job descriptions than the language typically preferred by male-coded algorithms. By applying bias mitigation to word embedding techniques, we need to develop a pre-processing layer that neutralizes these linguistic biases before they enter the decision-making model.

6.2 Work Package 2: Develop Auditing Toolkit

Currently, our auditing pipeline demonstrated, requires skills in programming and data science. This is a technical barrier and therefore excludes stakeholders like civil society organizations, trade unions, and non-technical external auditors, who are most invested in fair hiring mechanisms. To address this, the second work package should focus on the development of a no-code auditing tool.

This could be realized as a web interface that allows non-technical users to upload anonymized datasets, select sensitive attributes like gender and age, and execute the intersectional pipeline defined in our experiment. The tool will automatically generate a standardized fairness report that visualizes contrasting impact metrics and highlights potential cumulative penalty effects. This development is crucial for democratizing the auditing process, enabling external actors to verify compliance with the EU AI Act without relying on the technical resources of the audited companies.

6.3 Work Package 3: Human-Centric Validation and User Studies

Algorithmic fairness must be evaluated in the context of human decision-making processes. Even an algorithm that is mathematically fair can be misused if the human recruiter does not trust or understand its output. Therefore, the third work package should conduct a controlled user study with HR professionals to evaluate the quality of human oversight.

Participants could be divided into three groups:

- **Group A:** uses a standard black-box AI tool.
- **Group B:** uses a tool with SHAP explanations.
- **Group C:** uses our proposed tool with intersectional mitigation.

The study aims to measure whether explainability features actually assist recruiters in identifying bias or if they lead to information overload. Furthermore, we could investigate the phenomenon of automation bias, specifically whether a label "mitigated" causes recruiters to reduce their own critical evaluation of the systems decisions.

6.4 Project Timeline

As this is a research proposal, we came up with a timeline over a period of 24 months to ensure a realistic implementation:

- Months 1-6: Data Collection. We will gather real-world anonymized CVs and develop the text-analysis features from work package 1.
- Months 7-12: Software Development. We will build the prototype of the no-code web application so that people without technical background can test it.
- Months 13-18: User Testing. We will observe HR managers and work with trade unions to test the developed tool and application in practice.
- Months 19-24: Final Report. We will analyze the results and, most importantly, publish the open-source software and a policy guide for the EU.

7 Policy Frameworks

Technical tools alone are not enough to make AI systems fair and accountable. They need to be supported by clear institutional rules and legal frameworks. As demonstrated in our experiment, bias often conceals itself within the intersections of data, remaining invisible to standard metrics. Consequently, technical mitigation strategies must be embedded in governance structures.

7.1 Corporate Responsibility

Companies that use automated hiring tools should be required to carry out regular fairness checks, including bias measurements and transparency reports. These audits could become part of existing responsibility programs or mandatory ethical standards. To ensure genuine accountability, these internal checks must be validated by external oversight.

7.2 Institutional Response

At the policy level, regulators could set up systems with independent experts who are allowed to test, inspect, and certify AI models for fairness and transparency. This would make fairness not just a simple goal but a legal requirement.

To operationalize this independent testing, we propose a standardized auditing protocol for regulators:

- **Step 1 - Transparency:** Many commercial hiring tools are not publicly available. Auditors should be allowed to train interpretable models, like the one used in our case study, on representative data to approximate the decision boundaries of the private hiring system.
- **Step 2 - Diagnosis:** Institutional standards must conduct intersectional tests. Auditors must specifically probe for critical combinations such as gender and age to detect cumulative disadvantages.

- **Step 3 - Mitigation:** If bias is detected, the regulatory response should require the deployment of mitigation processes such as the *ThresholdOptimizer* to mathematically guarantee demographic parity.

7.3 Governance and the European Union AI Act

The European Union AI Act, which already treats hiring algorithms as high risk systems, offers a strong starting point for such governance structures. It requires to implement an effective human oversight and risk management systems. Our proposed auditing pipeline provides the technical means to fulfill these legal obligations.

7.4 The Role of Civil Society

Public engagement is also important. Collaborations between companies, universities, and civilians can help make AI development more transparent. Equality commissions and digital rights organizations can act as monitors to ensure that AI tools respect fundamental human rights.

Effective monitoring requires access to the right tools. By using public libraries like SHAP and Fairlearn, advocacy groups can support their ethical arguments with data. Instead of simply criticizing a system as opaque, they can produce clear evidence of unfairness and demand concrete technical improvements.

7.5 Education and Future Outlook

Education campaigns could also raise awareness among employers and job seekers about the risks and limits of automated hiring systems. Looking forward, stronger links between researchers, politicians, and data scientists are needed to make sure that fairness in AI is continuously monitored and improved. International cooperation can help create a common understanding of what ethical AI means. Automated hiring systems can become both effective and fair with clear rules, independent monitoring and public involvement.

8 Conclusion

Bias in automated hiring shows the ongoing challenges between technological progress and social fairness. As more organizations use algorithms to make important decisions, the principles of equality, transparency, and accountability need to be part of every step in the system design and use.

This research highlights that following general fairness principles is not enough. The most dangerous forms of discrimination are often hidden at the intersections of identity which remain invisible to standard auditing methods. XAI and fairness metrics are useful for reaching these goals, but their real value lies in their application within a rigorous auditing pipeline.

By combining tools like SHAP with intersectional mitigation strategies from Fairlearn, this project shows how technical methods can directly support human rights. Our experiment proved that it is technically feasible to detect invisible intersectional bias and remove it, restoring equal opportunity for marginalized subgroups.

While our experiment successfully demonstrated mitigation on tabular data, this should be only the first step. Looking forward, the complexity of auditing will increase as recruitment

shifts towards unstructured data and LLMs. Therefore, the auditing pipeline proposed here serves as a blueprint for the future research outlined in our agenda.

Consequently, algorithmic fairness is not just a technical problem. It is also a moral responsibility, strategically necessary to prevent the systematic loss of qualified talent, and a way to make sure that digital hiring systems do not repeat old patterns of inequality. The auditing protocol proposed in this paper offers a concrete path forward for regulators and civil society. With transparent models, intersectional auditing, and clear policy rules, AI can move from being a cause of bias to becoming a tool for fairness and equal opportunity.

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